POLYTECH NICE SOPHIA

Electronics and Embedded Systems Department

End of Fifth Year Studies Internship Report

3D Sonar Mapping, Surface Reflectivity and Real-Time SLAM Loop Closure

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Academic Year 2025 – 2026

Acknowledgements

I would like to express my gratitude to Prof. Yvan Petillot, Professor of Robotics and Autonomous Systems at Heriot-Watt University, member of the Ocean Systems Laboratory (OSL), and Deputy Director of the Institute for Sensor Signals and Systems (ISSS), for the opportunity to conduct my internship within the Ocean Systems Laboratory. His guidance and support were valuable throughout this work. I am thankful for the opportunity to work on research projects in 3D acoustic mapping and real-time SLAM.

I express sincere thanks to my internship tutor, Mr. Simon Archieri, Research Engineer at the Ocean Systems Laboratory, for his technical mentorship and regular feedback throughout this internship. His knowledge of underwater sensors, software engineering, and experimental methodology supported my work and engineering progression.

I would also like to thank Mr. Pascal Masson, Lecturer-Researcher at Polytech Nice Sophia, my academic referent, for his academic supervision and coordination during my fifth-year studies.

Thanks are due to the members, researchers, and PhD students of the Ocean Systems Laboratory and the Edinburgh Centre for Robotics for their support and technical discussions. Access to marine sensors, GPU computing infrastructure, and experimental underwater datasets was essential to this project.

Finally, I thank the faculty and staff of Polytech Nice Sophia (Electronics and Embedded Systems Department) for providing my engineering education over the past five years.

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Chapter 1

Introduction

1.1 Technical Context: 3D Acoustic Imaging

Underwater autonomous systems, including Autonomous Underwater Vehicles (AUVs) and Remotely Operated Vehicles (ROVs), are used in subsea inspection, offshore energy infrastructure monitoring, bathymetric mapping, environmental surveillance, and marine archaeology. In terrestrial and aerial robotics, optical cameras and LiDAR provide dense spatial and visual perception. However, underwater environments present sensing challenges for optical systems. Light experiences exponential attenuation, absorption, and backscattering caused by suspended particles and water turbidity. Consequently, optical imaging systems are restricted to short ranges of a few meters and become ineffective in turbid waters.

To overcome the physical limitations of light transmission in seawater, acoustic sensing serves as the primary perception modality for underwater mapping. Sound waves propagate with lower attenuation in water, allowing acoustic sensors to inspect subsea structures over tens of meters regardless of water turbidity or ambient lighting. Traditional acoustic sensors, such as single-beam echosounders, 2D sidescan sonars (SSS), and 2D multibeam forward-looking sonars (FLS), provide bathymetric profiles and 2D acoustic intensity images. However, 2D acoustic images lack explicit vertical geometry, creating ambiguities when mapping 3D structures such as shipwrecks, harbor walls, and underwater terrain.

The emergence of compact 3D acoustic imaging sensors, such as the Water Linked Sonar 3D-15, provides new capabilities for underwater perception. Operating at update rates from 5 Hz to 20 Hz with a wide field of view, this sensor directly outputs 3D point cloud measurements up to a range of 15 meters. This capability enables real-time 3D spatial mapping and structural modeling in marine environments.

1.2 Problem Statement: Subsea Navigation Drift

Operating autonomous vehicles in underwater environments presents navigation challenges. Electromagnetic signals, including GPS, cannot penetrate seawater. Underwater vehicles must therefore rely on dead-reckoning navigation systems combining Inertial Navigation Systems (INS), Doppler Velocity Logs (DVL), electronic compasses, and depth sensors.

While INS/DVL packages offer short-term accuracy, dead-reckoning navigation is subject to cumulative integration error. Drift in vehicle pose estimation degrades the consistency of the constructed 3D map. When an AUV revisits an area after an extended trajectory, drift causes misalignment between overlapping point clouds. In volumetric mapping, this discrepancy produces duplicate surfaces, phantom structures ("ghosting"), and geometric distortion, limiting

the map's utility for navigation and inspection.

To mitigate dead-reckoning drift and maintain spatial consistency, subsea mapping systems employ Simultaneous Localization and Mapping (SLAM) with Global Loop Closure Detection. The loop closure pipeline detects when the vehicle re-enters a previously surveyed region, establishes 3D point correspondences under acoustic noise, estimates 6-Degrees-of-Freedom (6-DOF) relative rigid body transformations, and incorporates these constraints into factor graph optimizations (e.g., GTSAM / iSAM2) to correct vehicle poses and the global 3D map.

1.3 Internship Objectives and Contributions

This fifth-year engineering internship was conducted within the Ocean Systems Laboratory (OSL) at Heriot-Watt University (Edinburgh Centre for Robotics), under the supervision of Prof. Yvan Petillot and technical mentorship of Mr. Simon Archieri. The primary objective was to implement 3D acoustic point cloud processing, surface reflectivity modeling, and global loop closure algorithms for real-time 3D sonar SLAM.

The technical contributions of this internship are summarized as follows:

1. 3D Acoustic Reflectivity Map Reconstruction (Project 1): Implemented a backscatter intensity correction model combining OpenVDB and NanoVDB Signed Distance Functions (SDF) with GPU ray-tracing and 3D Sobel normal estimation to isolate acoustic reflectivity from 3D sonar intensity measurements.
2. Real-Time Global Loop Closure Pipeline for 3D Sonar SLAM (Project 2): Developed an end-to-end, GPU-accelerated global loop closure framework for 3D point clouds. The pipeline integrates NanoVDB SDF ray-casting filters, 3D normal estimation, keypoint extraction (Harris3D, ISS), descriptor benchmarking (FPFH, SHOT, HPNVD, M2DP [16], NDTMC), and transformation solvers (Umeyama, RANSAC, TEASER++).
3. High-Performance C++/CUDA Architecture Refactoring: Implemented CUDA kernels for leaf building, radius searching, feature score computation, and descriptor extraction to enable high-throughput submap processing in real time.
4. Experimental Validation and Workshop Poster: Evaluated the pipelines on underwater datasets (quarry surveys and wave tank tests) and prepared a research poster titled "Global Loop Closure for Real-Time 3D Sonar SLAM", presented at the Marine Robotics Workshop (May 2026).

Chapter 2

Host Institution and Laboratory

2.1 Heriot-Watt University and ECR

Heriot-Watt University, established in 1821 in Edinburgh, conducts engineering and signal processing research within the Institute of Sensors, Signals and Systems (ISSS). In partnership with the University of Edinburgh, it hosts the Edinburgh Centre for Robotics (ECR) and The National Robotarium, supporting research and deployment of autonomous systems in subsea and hazardous environments.

2.2 The Ocean Systems Laboratory (OSL)

The Ocean Systems Laboratory (OSL) is a multidisciplinary research centre at Heriot-Watt University, integrated with the ECR and The National Robotarium. Led by Prof. Yvan Petillot, the OSL brings together academics, postdoctoral researchers, research engineers, and PhD candidates focusing on sensing, localization, mapping, and planning in underwater environments.

2.2.1 Core Research Pillars

The OSL focuses on four main research axes:

- Acoustic Sensing and Signal Processing: Ray tracing, acoustic modeling, and processing for 2D/3D imaging sonars.
- Subsea SLAM and Volumetric Mapping: Real-time implicit mapping (TSDF/SDF with OpenVDB/NanoVDB) and factor graph optimization (GTSAM/iSAM2) for GPS-denied navigation.
- Autonomous Control and Learning: Model Predictive Path Integral (MPPI) control and Reinforcement Learning for subsea intervention and multi-agent coordination.
- Shared Autonomy: Teleoperation interfaces and mission planning for offshore inspection.

2.2.2 Experimental Facilities

The laboratory maintains experimental and computing resources:

- In-House Water Tanks: Testing pools equipped with acoustic transceivers and optical motion tracking systems for hydrodynamic testing and sensor calibration.
- Field Robotics Fleet: Deployable Autonomous Underwater Vehicles (AUVs) and Remotely Operated Vehicles (ROVs) equipped with navigation sensors (DVL, FOG/INS) and acoustic

sensors (including the Water Linked Sonar 3D-15).

- GPU Computing Infrastructure: CUDA computing workstations and clusters for simulation (e.g., Isaac Sim marine environments) and volumetric mapping pipelines.

Chapter 3

Theoretical Foundations

3.1 3D Sonar SLAM System Overview

Autonomous underwater navigation and 3D subsea mapping require handling acoustic point clouds while maintaining spatial consistency along survey trajectories. The framework developed and extended during this internship is a modular, GPU-accelerated C++/CUDA and ROS 2 SLAM framework tailored for 3D acoustic multibeam imaging sonars.

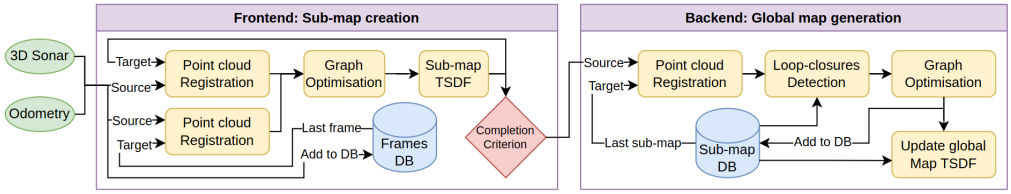


Figure 3.1: End-to-end architecture of the `roller` 3D Sonar SLAM framework.

As illustrated in Figure 3.1, the system is architected around a clear decoupling between two execution tiers:

Frontend: Submap Creation and Local Tracking: The frontend operates synchronously at sensor rate to perform high-rate frame registration, local keyframe graph optimization, and volumetric submap construction:

- Sensor Ingestion and Pre-processing: Ingests sonar point clouds (x, y, z, I) alongside odometry (DVL) and AHRS/INS attitude and depth measurements. Low-density pings are discarded.
- Keyframe Point Cloud Registration: Surface and normal vectors are extracted for incoming keyframes. Point-to-plane scan-matching aligns the current frame against both the previous keyframe and the active submap, storing verified frames in the frames database.
- Local Keyframe Graph Optimization: A local iSAM2 instance optimizes keyframe poses $(k_0, \ldots, k_n)$ within the current submap, ensuring locally consistent trajectory.
- Completion Criterion and Volumetric TSDF Fusion: A submap completion trigger monitors keyframe accumulation, finalizing the submap when a maximum keyframe count is reached or when the first keyframe of the submap isn't visible anymore. Upon completion, accumulated keyframes are integrated into a GPU-accelerated volumetric Truncated Signed Distance Function (TSDF) sparse voxel grid (NanoVDB). Observation weight thresholding eliminates acoustic noise and artifacts, followed by Marching Cubes isosurface extraction and analytical

3D spatial gradient normal computation. The completed submap is packaged and handed off to the backend.

Backend: Global Optimization and Map Generation: The backend operates asynchronously at the submap timescale to maintain map consistency:

- Submap Storage and Inter-Submap Registration: The completed submap is inserted into the global submap database (Submap DB) as a new pose node S_i. Point cloud registration aligns the new submap against adjacent submaps to introduce inter-submap odometry constraints.
- Loop Closure Detection: Global and local feature descriptors evaluate candidate matches between the new submap and historical submaps in the database to detect revisit events.
- Global Graph Optimization: A global iSAM2 factor graph jointly optimizes all submap poses, fusing inter-submap scan-matching constraints, loop closure factors, and drift-free AHRS orientation and hydrostatic depth priors.
- Global TSDF Map Update and Dense Outputs: Optimal graph corrections are applied across the submap database. In an asynchronous background thread, corrected submaps are integrated into a global volumetric TSDF representation to publish dense 3D point clouds, generate 3D OctoMap occupancy grids, and export benchmark trajectory logs.

3.2 3D Acoustic Sonar Physics

Unlike optical signals which experience exponential absorption and scattering in turbid water, acoustic waves propagate with lower attenuation in seawater. This allows acoustic multibeam sonars to perceive and map 3D underwater environments across ranges where optical cameras and LiDAR cannot operate.

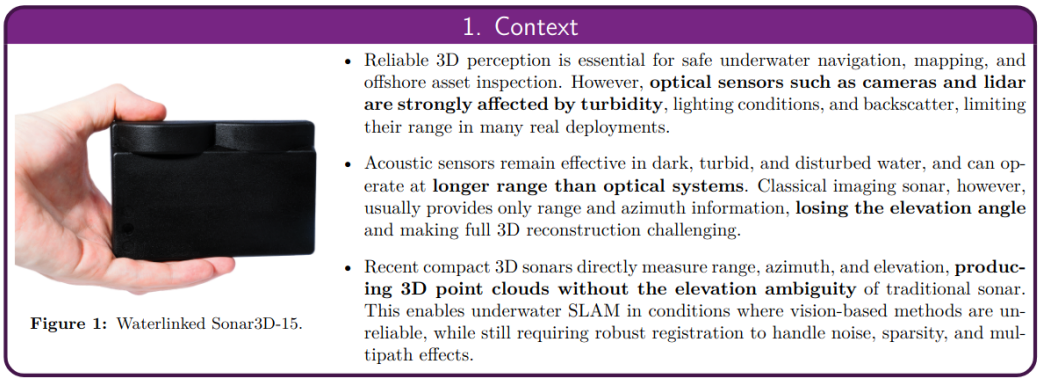


Figure 3.2: The Water Linked Sonar 3D-15 sensor and its subsea operational context.

The sensing instrument utilized in this work is the Water Linked Sonar 3D-15 (Figure 3.2), a compact phased-array 3D acoustic sensor delivering a 90° (horizontal) × 40° (vertical) field of view with an operating range from 0.20 to 15.0 m (at 5 Hz) and up to 4.0 m at 20 Hz, streaming 3D Cartesian point clouds (x, y, z) paired with acoustic intensity I over Ethernet.

Acoustic Noise and Measurement Artifacts: Unlike terrestrial LiDAR point clouds, 3D acoustic point clouds exhibit distinct noise modalities that must be addressed:

- Acoustic Speckle Noise: Interference among backscattered sound waves from micro-roughness on the seabed induces spatial noise on range measurements.

- Multipath Reflections: Sound waves bouncing between the seabed, the sea surface, and the vehicle hull generate ghost points behind or above physical structures.
- Non-Uniform Spatial Density: Beam divergence causes point density to decrease quadratically with distance ($1/R^2$), creating dense point clusters near the sensor and sparser measurements at longer ranges.
- Grazing Angle Dropout: Acoustic energy reflected from surfaces oriented nearly parallel to the beam direction ($\theta \to 90°$) fails to backscatter toward the receiver, leading to dropouts on steep slopes.

3.3 Volumetric 3D Representations

To convert unorganized acoustic point clouds into continuous 3D surfaces, the system employs an implicit volumetric surface representation based on Truncated Signed Distance Functions (TSDF).

3.3.1 TSDF and Volumetric Fusion

To represent 3D underwater environments, the spatial domain is discretized into a regular grid of cubic volume elements called voxels. Instead of storing binary occupancy, the Signed Distance Function (SDF) assigns each voxel a value representing its shortest distance to the nearest physical surface:

- Positive values indicate empty space in the water column (in front of the surface).
- Zero-crossing corresponds to the physical boundary of the seabed or structure.
- Negative values indicate regions inside solid matter (behind the surface).

To optimize memory usage and focus computational resources on the mapped terrain, distance values are truncated to a margin around the surface, forming the Truncated Signed Distance Function (TSDF).

When successive 3D sonar frames are acquired, incoming distance measurements and confidence weights are merged into the voxel grid using a running weighted average formulation [5]. Observed surfaces accumulate confidence weights over time, converging to consistent geometric surfaces, while transient acoustic noise, phantom points, and multipath reflections average out (Figure 3.3).

3.3.2 Sparse Grids: OpenVDB and NanoVDB

Standard dense 3D volumetric grids scale with $O(N^3)$ memory complexity, making them impractical for large-scale subsea mapping (a $50\,\text{m} \times 50\,\text{m} \times 10\,\text{m}$ volume at 5 cm resolution would require $> 200 \times 10^6$ voxels and gigabytes of memory).

To achieve memory and computational efficiency, the `roller` pipeline implements sparse hierarchical tree structures based on OpenVDB [6] on the CPU host and NanoVDB [7] on the GPU device:

- Tree Topology ($B_1 \times B_2 \times B_3$): A 4-level tree consisting of a Root Node (dynamic map), Internal Nodes (32^3 and 16^3), and Leaf Nodes ($8^3 = 512$ voxels). Unobserved regions in the water column allocate zero memory.
- NanoVDB Linearized Structure: OpenVDB trees are serialized into a contiguous byte buffer with relative offset addressing. This pointerless layout allows parallel ray-tracing, zero-crossing surface extraction, and 3D Sobel filtering directly inside CUDA thread blocks without CPU-GPU memory divergence.

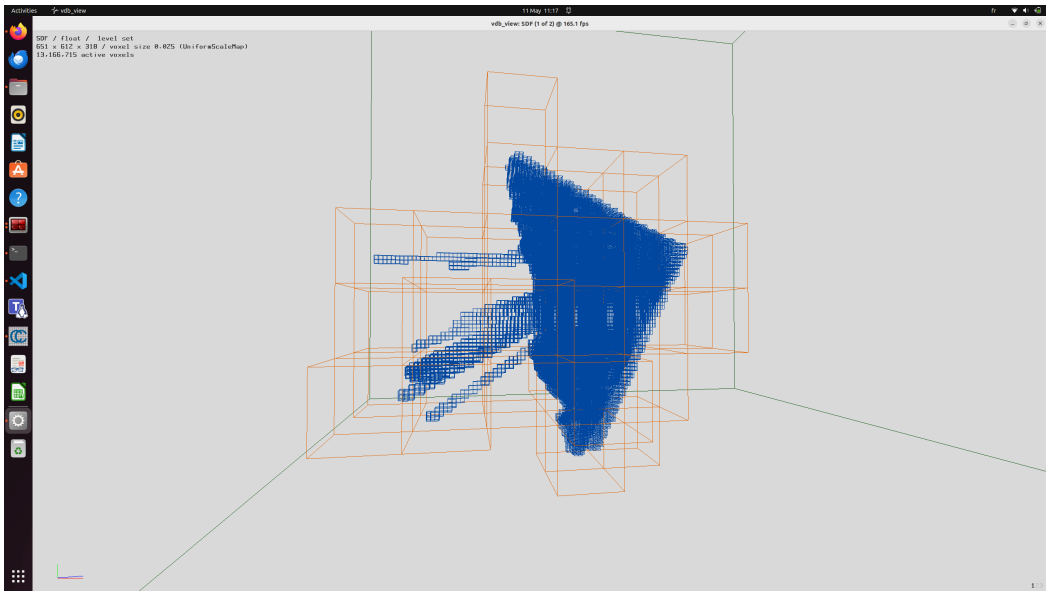


Figure 3.3: Volumetric TSDF voxel grid occupancy. Visualization of the allocated voxel grid structure representing the mapped underwater surface geometry.

3.3.3 Surface Normal Estimation via 3D Sobel SDF

A benefit of implicit Signed Distance Fields is that surface normals can be extracted analytically from the spatial gradient of the SDF field, bypassing point-cloud nearest-neighbor searches.

Because the distance field changes along the direction perpendicular to the surface boundary, a 3D Sobel filter evaluates the spatial derivatives of distance values along the x, y, and z axes across neighboring voxels. Normalizing this spatial gradient yields outward-pointing unit normal vectors directly on the GPU.

3.4 Underwater SLAM and Factor Graph Optimization

Dead-reckoning navigation in GPS-denied underwater environments accumulates integration drift over time. To maintain trajectory and map consistency, subsea SLAM frameworks formulate trajectory estimation as a Factor Graph Optimization problem [8]. Unlike classical Extended Kalman Filters (EKF-SLAM) that linearize non-linear models only once and discard past poses, factor graph smoothing preserves the full trajectory history. This formulation avoids linearization errors under motion drift, enables retroactive trajectory correction upon loop closure, and exploits matrix sparsity for real-time incremental solving via iSAM2.

3.4.1 Factor Graph Trajectory Estimation

A factor graph is a bipartite graphical network that models the SLAM problem using two types of connected nodes (Figure 3.4):

- State Variables (Poses): Represent the estimated 3D position and orientation (6-DOF pose) of the vehicle at successive time steps along its trajectory.
- Factors (Constraints): Represent probabilistic constraints derived from sensor measurements, each weighted by its measurement confidence or uncertainty.

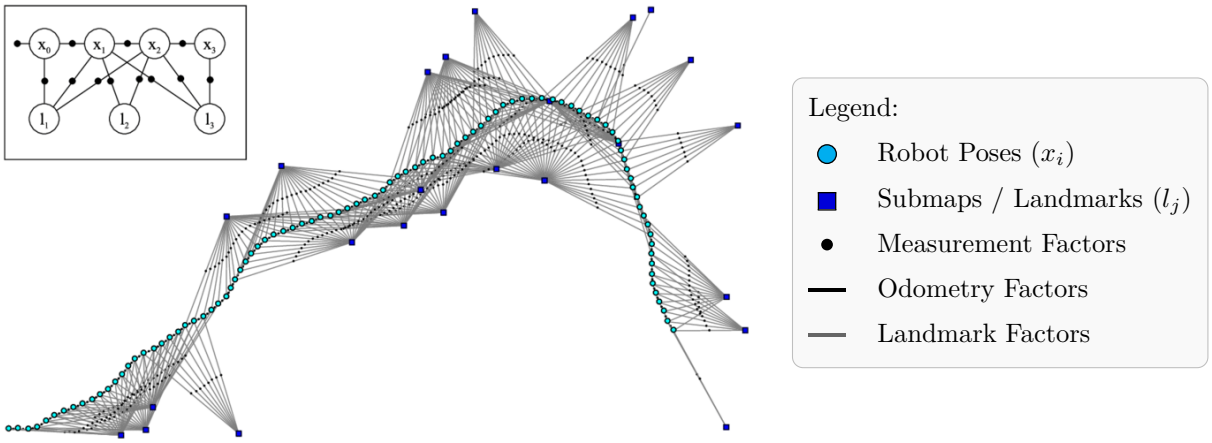


Figure 3.4: Factor graph formulation for simultaneous localization and mapping (SLAM).

The goal of the optimizer is to determine the trajectory that best satisfies factor constraints, finding the configuration that minimizes the overall error between expected and measured values across the survey.

3.4.2 Factor Types in the SLAM Backend

The factor graph topology integrates three distinct categories of factors:

1. Prior Factors: Anchor the vehicle's initial starting pose in the global coordinate frame with fixed uncertainty, preventing the map from floating unconstrained.
2. Odometry Factors: Connect consecutive poses based on dead-reckoning navigation (DVL velocity, INS orientation, and depth) coupled with high-rate scan-matching (GICP) [10]. These factors measure relative step-by-step displacements between adjacent poses.
3. Global Loop Closure Factors: Created whenever the global loop closure backend detects and geometrically verifies that the current submap overlaps with a previously visited submap. By introducing a direct geometric constraint between non-consecutive poses, loop closure factors pull drifting trajectory segments back into alignment and eliminate accumulated drift.

To perform optimization in real time, the framework utilizes iSAM2 [9] (Incremental Smoothing and Mapping with the Bayes Tree) within the GTSAM library [8]. Instead of resolving the entire historical graph upon each measurement, iSAM2 updates only the subset of trajectory variables directly affected by new observations and loop closures, achieving millisecond-scale execution times.

3.5 3D Feature Extraction and Descriptors

Establishing 6-DOF loop closure constraints across 3D acoustic submaps requires feature detection and descriptive geometric encoding. The feature extraction pipeline is structured in three stages: keypoint detection, local descriptor encoding, and global place recognition.

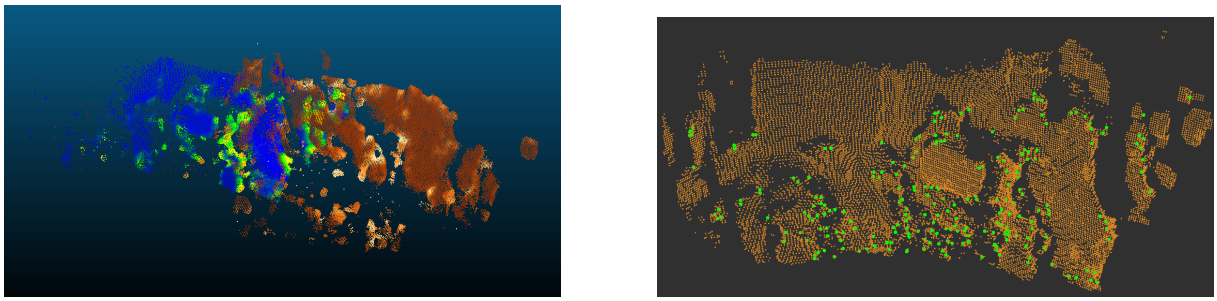
3.5.1 Keypoint Detectors

Directly matching all voxels in a submap is computationally expensive and prone to ambiguity in planar seabed regions. Keypoint detectors isolate points exhibiting local geometric distinctiveness.

3.5.1.1 3D Harris Keypoint Detector

The 3D Harris operator [12] extends the 2D image corner detector [11] to 3D surface meshes and point clouds. It identifies geometric landmarks (such as corners, vertices, and structural boundaries) by analyzing how surface normals vary within a local spherical neighborhood.

If the surface normals across the neighborhood point in multiple directions, the surface exhibits multi-directional curvature rather than being planar. Points with the highest corner-like response values correspond to salient features on the seabed (Figure 3.5(a)).



(a) Continuous 3D Harris response field

(b) Salient keypoints after 3D NMS

Figure 3.5: 3D Harris keypoint extraction on acoustic bathymetry: (a) the continuous 3D structure tensor response field R_{Harris} highlights regions of high surface curvature, and (b) spatial Non-Maximum Suppression (NMS) isolates well-spaced, repeatable keypoints on seabed structures.

3.5.1.2 Intrinsic Shape Signatures (ISS)

The Intrinsic Shape Signature (ISS) detector [13] identifies keypoints by analyzing how neighboring 3D points are spatially distributed around each candidate point along its principal geometric axes (the directions of maximum, medium, and minimum spread).

A point is classified as a keypoint if its neighborhood exhibits variation in all three spatial dimensions. This criterion filters out uninformative points located on flat planar seabeds (which spread in only two directions) or along straight, rotationally symmetric ridges, selecting keypoints that possess 3D shape characteristics (Figure 3.6).

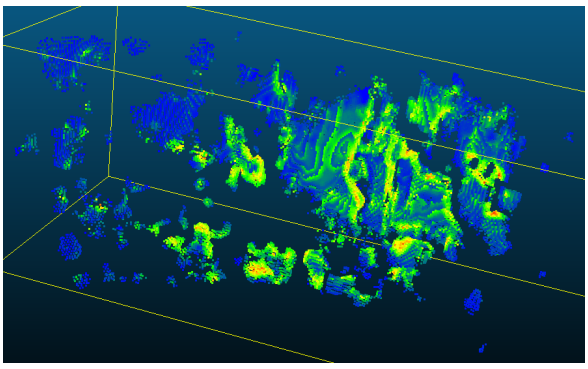


Figure 3.6: Intrinsic Shape Signature (ISS 3D) saliency score map evaluated across the submap point cloud, highlighting distinctive 3D surface variations.

3.5.1.3 Tensor Saliency Keypoint Estimation and PCA

Local surface geometry can also be characterized using Principal Component Analysis (PCA). For each point and its local neighborhood, PCA fits an oriented 3D bounding ellipsoid by identifying

the principal directions of spatial spread. The resulting eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq 0$ describe the semi-axis lengths along each principal direction, normalized by their total sum such that $\lambda_1 + \lambda_2 + \lambda_3 = 1$. To quantify local 3D structural distinctiveness while suppressing planar seabeds, keypoints are scored using omnivariance:

$$O = \sqrt[3]{\lambda_1 \lambda_2 \lambda_3} \tag{3.1}$$

Omnivariance measures the geometric mean of the normalized principal spreads, attaining higher values on 3D corners, rocks, and junctions while dropping toward zero on flat planar terrain (Figure 3.7). Furthermore, to suppress acoustic speckle noise, PCA estimation is evaluated across multiple neighborhood radius scales, aggregating multi-scale omnivariance responses.

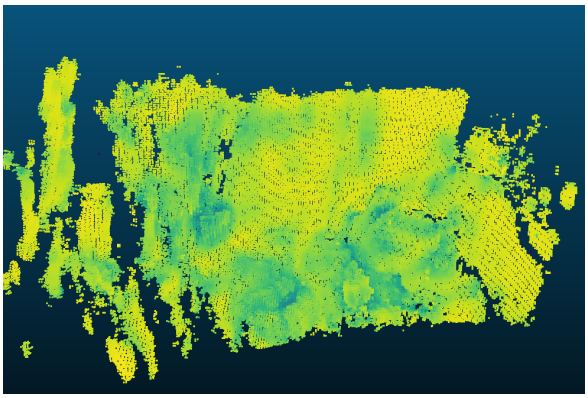


Figure 3.7: Tensor saliency score computation and keypoint detection. Visualization of the continuous 3D tensor saliency response across the submap geometry and the resulting salient keypoints extracted on distinctive surface structures.

3.5.1.4 Spatial Non-Maximum Suppression (NMS)

To prevent keypoints from clustering on a single edge or corner, a 3D Non-Maximum Suppression (NMS) filter retains only the candidate point with the highest saliency response within a given radius. This provides spatial dispersion of keypoints across the submap (Figure 3.5(b)).

3.5.2 Local Feature Descriptors: FPFH and SHOT

Local feature descriptors encode the local surface geometry surrounding each keypoint into rotation-invariant feature vectors. Instead of matching raw point coordinates—which depend on the vehicle's position—local descriptors capture geometric properties such as curvature, relative surface orientation, and spatial point distributions within a local neighborhood. This enables point-to-point correspondence matching between overlapping submaps.

3.5.2.1 Fast Point Feature Histograms (FPFH)

Fast Point Feature Histograms (FPFH) [14] characterize local surface geometry by analyzing how surface normals deviate relative to each other within a spherical support radius around the keypoint.

The method computes angular differences between the keypoint's normal and each neighboring normal. These angular relationships capture local curvature. The angles are binned into localized histograms and aggregated with neighbor histograms using distance-weighted averaging. This aggregation allows FPFH to encode surface geometry while remaining computationally lightweight.

3.5.2.2 Signature of Histograms of OrienTations (SHOT)

The Signature of Histograms of OrienTations (SHOT) descriptor [15] generates a signature by jointly capturing spatial point distributions and surface orientation. It begins by establishing a local reference frame centered on the keypoint.

The spherical volume around the keypoint is divided into a grid of spatial sectors (partitioned into radial shells, elevation bands, and azimuthal slices). Within each sector, a histogram records orientation differences between local surface normals and the central keypoint normal. Concatenating histograms from all sectors yields a descriptor encoding spatial layout and surface orientation variations.

3.5.3 Global Descriptors: STD, NDTMC and ESF

While local descriptors enable point-to-point correspondence matching, global descriptors compress the 3D geometry of a submap into a compact signature for place recognition and submap candidate retrieval.

3.5.3.1 Stable Triangle Descriptor (STD)

The Stable Triangle Descriptor (STD) [17] represents 3D submaps through geometric triangles formed by triplets of keypoints located within a specified metric distance range.

To achieve rotation and viewpoint invariance, each triangle is characterized by its three sorted side lengths and the relative orientation angles of the surface normals at its vertices. These triangle signatures are indexed into a multi-key hash table, enabling retrieval of candidate submap matches during global loop closure searches.

3.5.3.2 Normal Distributions Transform Map Code (NDTMC)

The Normal Distributions Transform Map Code (NDTMC) [18] summarizes the 3D geometry of a submap by dividing the volume into regular grid cells. Within each cell, the local point cloud is modeled as a 3D Gaussian distribution, capturing both its spatial center and local geometric orientation.

These structural features, together with point density and elevation profiles, are projected onto a 2D multi-layer cylindrical grid to produce a fixed-size global matrix descriptor. Similarity between submaps is evaluated via circular cross-correlation, enabling heading-invariant place recognition and candidate retrieval.

3.5.3.3 Ensemble of Shape Functions (ESF)

The Ensemble of Shape Functions (ESF) [19] is a global 3D descriptor that captures the structural and volumetric morphology of a submap without requiring surface normal estimation or local reference frames.

ESF constructs a 640-dimensional signature by aggregating probability histograms over randomly sampled point pairs and triplets, evaluating point-to-point Euclidean distances ($D2$), triangle interior angles ($A3$), triangle areas ($D3$), and distance ratios ($D2$ Ratio). By classifying line segments passing through terrain volume versus exterior open space, ESF provides SE(3) invariance and resilience to acoustic noise and sparsity for candidate retrieval.

Chapter 4

Work Realized

4.1 Project 1: Acoustic Reflectivity Reconstruction

The first objective of this internship was to implement a processing pipeline for reconstructing 3D acoustic reflectivity maps of underwater environments.

While the Water Linked Sonar 3D-15 produces 3D point cloud measurements accompanied by raw acoustic intensity values $I \in [0, 255]$, these raw values cannot be directly interpreted as physical surface properties. In this project, a GPU-accelerated processing pipeline was developed within the `roller` SLAM framework to isolate the acoustic reflectivity of subsea structures and seabed materials.

4.1.1 Physical Backscatter Modeling and Angle Correction

The reflectivity reconstruction pipeline is based on acoustic wave propagation principles [2, 3] and the backscatter model established by Coiras, Petillot, and Bell [4]. In Section B of their work, the acoustic backscatter intensity equation (Formula 1) models the observed echo intensity I recorded by the sonar receiver as:

$$I = \frac{K}{r^2} \cdot R \cdot \cos^n(\theta) \tag{4.1}$$

where:

- K is an aggregated calibration factor encompassing transmitted acoustic power, transducer beam patterns, and receiver gains.
- r is the slant range from the transducer to the insonified target point, capturing geometric acoustic transmission loss ($1/r^2$ spherical spreading).
- θ is the local angle of incidence between the incoming acoustic beam direction s and the local surface normal n ($\cos\theta = n \cdot s$).
- $\cos^n(\theta)$ represents the generalized Lambertian backscattering law (with empirical roughness exponent $n \geq 1$), characterizing how surface micro-roughness scatters sound energy back toward the transducer.
- R is the intrinsic surface acoustic reflectivity (backscatter coefficient), determined by the acoustic impedance mismatch between seawater and the target material.

In 2D side-scan sonar imaging [4], 3D elevation is unobserved, requiring shape-from-shading optimization to simultaneously estimate surface gradients and reflectivity. In contrast, with the

3D Sonar 3D-15 and real-time volumetric SLAM, the 3D spatial coordinate (x, y, z), the range r, and the surface normal n are determined by the GPU-accelerated TSDF model.

Consequently, we invert Formula (4.1) to compute the material reflectivity R for each incoming measurement:

$$R = \frac{I_{\text{normalized}}}{\max\left(\cos^n(\theta),\, \cos_{\min}\right)} \tag{4.2}$$

where a regularizing lower bound $\cos_{\min} \approx 0.1$ prevents numerical instability and noise amplification at grazing angles ($\theta \to 90°$). This inversion isolates acoustic reflectivity, making it invariant to the vehicle's viewpoint and distance.

4.1.2 GPU SDF Ray-Tracing and 3D Sobel Normals

A challenge in incidence angle correction is estimating reliable surface normals. Direct normal estimation from raw point clouds via nearest neighbors is affected by acoustic speckle and measurement noise.

To overcome this limitation, the pipeline extracts surface geometry directly from the implicit Signed Distance Function (SDF) generated during volumetric SLAM:

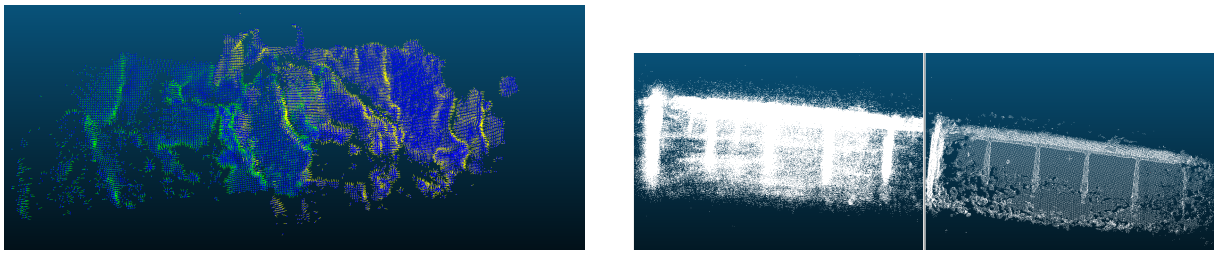
1. Volumetric Grid Preparation and Smoothing: The OpenVDB TSDF tree is processed via signed flood-fill and level-set Gaussian smoothing to obtain continuous distance gradients across the zero-crossing surface interface. The grid is then linearized into a NanoVDB buffer on the GPU.
2. Zero-Crossing Ray Tracing: For each acoustic return, a ray is cast from the known sensor pose toward the measurement. A GPU-accelerated zero-crossing algorithm identifies the intersection point with the reconstructed implicit surface boundary.
3. 3D Sobel Normal Extraction: At the zero-crossing voxel, a 3D Sobel operator spanning a $3 \times 3 \times 3$ voxel neighborhood computes the spatial gradient of the distance field. Normalizing this gradient provides analytical, outward-pointing unit normal vectors on the GPU (Figure 4.1a).

4.1.3 Voxel Filtering and Reflectivity Extraction

Following the incidence angle correction and normal estimation steps, the spatial measurements are integrated into a unified volumetric representation. Four separate sparse OpenVDB grids are updated concurrently: reflectivity, raw intensity, surface normals, and observation confidence weights. Each active surface voxel accumulates incoming observations across all survey submaps using a running weighted average.

To eliminate multipath artifacts and spurious reflections, voxels with insufficient observation hit counts, excessive distance from the zero-crossing, or unstable grazing angles are pruned from the active grid (Figure 4.1b).

Finally, the validated volumetric reflectivity grid is sampled to extract a 3D point cloud (x, y, z, R). This reconstructed reflectivity point cloud enables seabed materials (such as quarry rocks vs. waterlogged wood) to be compared based on their physical acoustic reflection properties.



(a) 3D Sobel surface normal extraction (b) TSDF ray-tracing noise filtering

Figure 4.1: GPU-accelerated geometric processing and volumetric filtering for reflectivity mapping: (a) analytical surface normals computed from the SDF distance gradient, and (b) point cloud denoising and multipath artifact removal via TSDF zero-crossing ray tracing.

4.2 Project 2: Real-Time Global Loop Closure

4.2.1 Project Overview: Subsea Loop Closure

Autonomous underwater navigation in GPS-denied environments relies on dead-reckoning sensors, including Doppler Velocity Logs (DVL), Inertial Navigation Systems (INS), and depth pressure sensors. While these instruments provide short-term motion tracking, integrating velocity and angular rate measurements over extended trajectories leads to cumulative error.

As the vehicle traverses a survey area, this drift causes the estimated vehicle trajectory to deviate from ground truth. When the vehicle revisits a previously surveyed site, the spatial mismatch between historical and current measurements manifests as mapping artifacts: duplicate structures ("double walls"), phantom obstacles, and geometric distortion.

A loop closure is the process of recognizing that the robot has returned to a previously mapped location and establishing a direct geometric constraint between the current pose and the historical pose (Figure 4.2). By inserting this constraint into a global optimization framework (factor graph smoothing), accumulated drift is corrected across the historical trajectory.

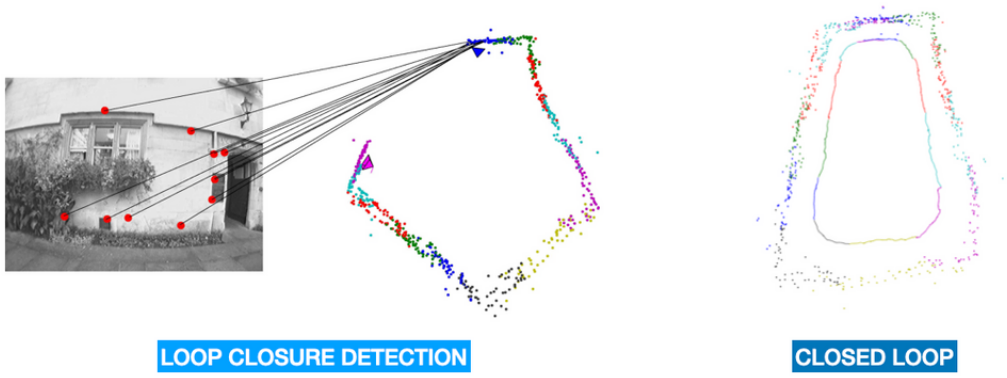


Figure 4.2: Principle of loop closure detection and drift correction in underwater SLAM.

4.2.2 Local vs. Global Loop Closure

In robotic mapping literature, a distinction is made between local loop closure (scan-to-scan / scan-to-submap tracking) and global loop closure (place recognition and global registration):

4.2.2.1 Local Scan-Matching (ICP)

Local registration frameworks (such as Iterative Closest Point, GICP [10], and Normal Distributions Transform) rely on iterative gradient-descent optimization to align consecutive point clouds. They operate under the assumption that an accurate initial estimate of relative displacement is available from dead-reckoning navigation.

While effective for frame-to-frame odometry (5–20 Hz), local scan-matching is insufficient for closing large survey loops in underwater environments due to its narrow basin of convergence. Iterative algorithms like ICP converge to the global alignment only if the initial pose error is within a narrow margin (typically $< 0.5\,\text{m}$ and $< 5°$). When an AUV navigates over extended distances, cumulative dead-reckoning drift exceeds several meters, causing local ICP to get trapped in local minima or diverge.

4.2.2.2 Global Place Recognition and Registration

In contrast, global loop closure operates without initial pose assumptions (360° rotation and translation invariance):

- It extracts viewpoint-invariant geometric descriptors (local descriptors like FPFH/SHOT or global descriptors like NDTMC/STD) to query the historical database and recognize revisited areas regardless of heading.
- It establishes feature correspondences globally and solves the 6-DOF relative transformation using outlier-resilient algorithms (such as TEASER++ and RANSAC), which can identify rigid-body transformations under high correspondence outlier ratios.

Global loop closure provides the geometric constraints needed to correct drift and maintain global consistency.

4.2.3 Initial CPU Benchmark on Synthetic Submaps

To evaluate and benchmark each stage of the registration pipeline without the confounding effects of live SLAM drift, a dedicated offline testing setup was established.

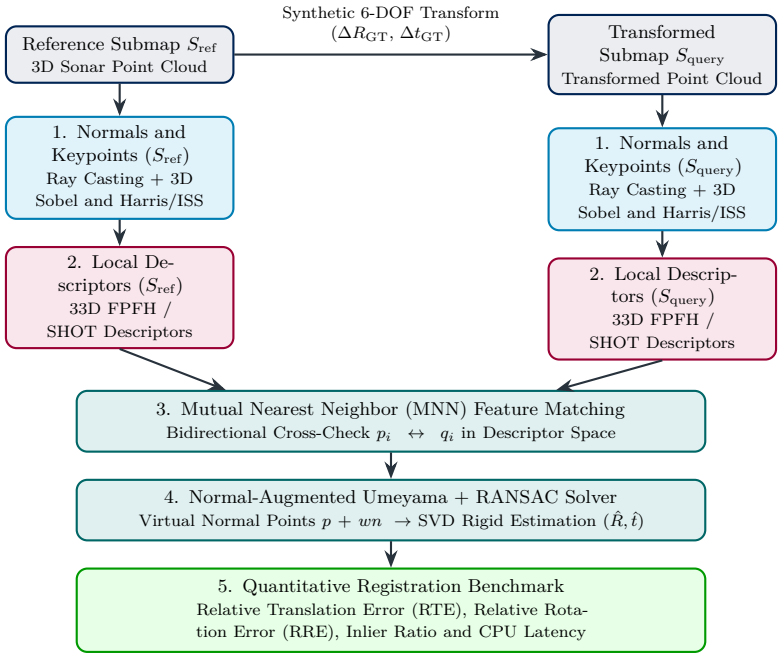


Figure 4.3: Workflow of the initial CPU-only loop closure benchmarking setup.

In this experimental setup (Figure 4.3):

- Controlled 6-DOF Ground Truth Transformation: A reference submap point cloud S_{ref} was duplicated and subjected to synthetic rigid-body transformations $(\Delta R_{\text{GT}}, \Delta t_{\text{GT}})$ spanning arbitrary yaw angles and metric translations.
- CPU Feature Extraction: Surface normals were extracted on CPU via volumetric ray casting through the TSDF voxel grid to locate isosurface zero-crossings, followed by a 27-neighborhood 3D Sobel gradient operator computing unit normals $n = -\nabla\phi/\|\nabla\phi\|$. Keypoints (Harris3D, ISS) and local feature descriptors (FPFH, SHOT) were subsequently extracted using CPU Point Cloud Library (PCL) implementations [25].
- Mutual Nearest Neighbor (MNN) Feature Matching: Correspondences were established by enforcing bidirectional cross-consistency between descriptor sets (MNN), discarding asymmetric one-way candidate matches.
- Normal-Augmented Umeyama with RANSAC: In underwater environments, planar seabeds and low-curvature structures can introduce rotational ambiguities if registration relies purely on 3D spatial positions. To constrain 6-DOF alignment, the Umeyama SVD algorithm [20] was extended to incorporate surface normals. For each correspondence pair (p_i, q_i) with unit normals (n_i, m_i), a secondary virtual point was created along the surface normal direction:

$$p'_i = p_i + w \cdot n_i, \qquad q'_i = q_i + w \cdot m_i \tag{4.3}$$

  where $w = 0.1\,\text{m}$ is a normal weighting factor. By concatenating the original points and the normal-offset points into $3 \times 2N$ measurement matrices:

$$P_{\text{aug}} = \begin{bmatrix} p_1 \dots p_N & p'_1 \dots p'_N \end{bmatrix}, \qquad Q_{\text{aug}} = \begin{bmatrix} q_1 \dots q_N & q'_1 \dots q'_N \end{bmatrix} \tag{4.4}$$

  the Umeyama SVD optimization simultaneously aligns spatial positions and surface normal orientations in closed form. Outliers were rejected using RANSAC sample consensus [21] (‘CorrespondenceRejectorSampleConsensus‘) enforcing both spatial and normal consistency.

This initial benchmark highlighted two limitations of the CPU baseline:

1. CPU Latency Bottleneck: While operating on submaps grants a larger computational time window than frame-to-frame tracking, the CPU pipeline incurred high latency. Specifically, normal calculation and keypoint extraction took several seconds per submap, creating a processing backlog during continuous autonomous navigation.
2. RANSAC Performance Under High Outlier Ratios: Due to acoustic speckle noise and perceptual aliasing in sonar data, the correspondence outlier ratio often exceeded 70–85%. Under high contamination, RANSAC required an exponential number of sample draws, causing runtime variations and occasional convergence to incorrect local alignments.

4.2.4 Integrated Real-Time GPU Framework

Building upon the insights gained from the initial offline tests, the complete loop closure pipeline was integrated directly into the live `roller` ROS 2 [26] SLAM framework with full GPU acceleration.

The integrated framework connects the real-time volumetric SLAM frontend to an asynchronous global loop closure backend, coordinating parallel surface geometry and acoustic intensity pre-processing, GPU-accelerated spatial filtering, saliency keypoint detection, multi-modal descriptor generation, multi-stage candidate retrieval, and 6-DOF registration. The complete dataflow is illustrated in Figure 4.4.

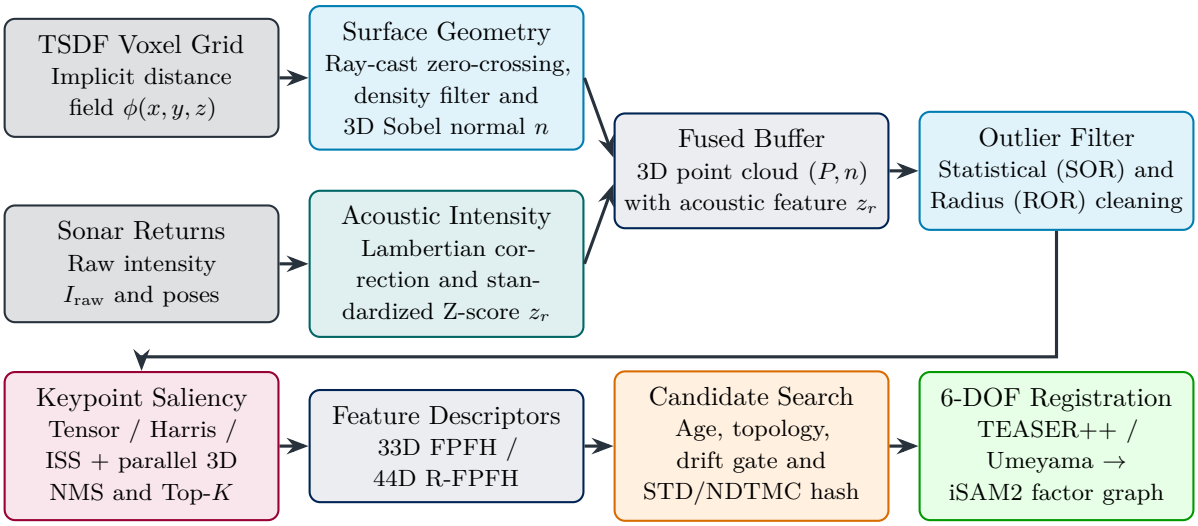


Figure 4.4: Architecture and dataflow of the SLAM-integrated 3D loop closure pipeline.

The integrated framework is organized into five core technical stages:

4.2.4.1 1. Parallel Geometry and Intensity Pre-Processing

When a submap is finalized by the SLAM frontend, it is processed through two parallel GPU pipelines:

- Branch A (Surface Geometry and Normals): The TSDF distance field $\phi(x, y, z)$ is queried to locate the implicit isosurface boundary via zero-crossing detection. A 26-neighborhood density filter discards floating acoustic noise. Central-difference spatial gradients $\nabla\phi$ are computed and normalized to obtain smooth, outward unit surface normals $n = -\nabla\phi/\|\nabla\phi\|$.
- Branch B (Acoustic Intensity Modeling and Z-Score Standardization): For each acoustic return, the ray vector $v = P_{\text{surface}} - P_{\text{sensor}}$ and incidence angle cosine $\cos\theta = n \cdot v$ are calculated. Physical Lambertian angle correction $r = I_{\text{raw}}/\max(\cos\theta, \epsilon)$ eliminates shading dependency, and intensity Z-score standardization $z_r = (r - \mu_r)/\sigma_r$ produces a normalized acoustic feature invariant to sensor gain variations.

The geometric positions, surface normals, and standardized acoustic intensities are fused into an initial surface point cloud buffer.

4.2.4.2 1.5. Surface Outlier Filtering (SOR and ROR)

To eliminate residual floating multi-path artifacts, two sequential filters are applied:

- Statistical Outlier Removal (SOR): Computes mean neighbor distances across $k = 50$ nearest neighbors and rejects points whose mean distance exceeds $\mu + \alpha\sigma$ (SOR).
- Radius Outlier Removal (ROR): Prunes isolated points with fewer than $M = 5$ neighbors within a search sphere of radius $R = 0.3\,\text{m}$ (ROR).

4.2.4.3 2. Keypoint Saliency Extraction and 3D NMS

Keypoint extraction evaluates local structural saliency using custom CUDA device kernels:

- Saliency Detectors: Keypoint candidates are scored via multi-scale Tensor Saliency (Omnivariance $Z = \sqrt[3]{\lambda_1\lambda_2\lambda_3}$), Harris 3D corner response ($Z = \det(M) - k\,\text{trace}(M)^2$), or Intrinsic Shape Signatures ($Z = \lambda_3/\lambda_1$).

- 3D Non-Maximum Suppression (NMS) and Top-K Filtering: Saliency peaks are extracted using a parallel 3D NMS radius filter, followed by a Top-K selection ($K = 2000$) to guarantee a well-distributed, repeatable set of keypoints across the submap.

4.2.4.4 3. Multi-Modal Feature Description

The pipeline extracts distinctive descriptors at each keypoint using modern C++ type-safe static polymorphism:

- Geometric FPFH (33D): Encodes spatial normal angular variations within a spherical support radius (FPFH).
- Acoustic R-FPFH (44D): Augments the 33D geometric descriptor with an 11-bin histogram of standardized acoustic reflectivity scores z_r (R-FPFH), allowing the matcher to distinguish geometrically identical structures that differ in material composition.

4.2.4.5 4. Candidate Submap Retrieval

To minimize unnecessary matching computations, candidate submap pairs (S_i, S_j) must pass a sequence of gating filters:

1. Temporal Exclusion: Ensures $|i - j| > \Delta_{\text{age}}$ to prevent trivial matching against immediate past submaps.
2. Factor Graph Topology Check: Skips pairs that are already constrained by existing factor graph edges.
3. Uncertainty-Aware Distance Gating: Verifies that the spatial distance is within $R_{\text{base}} + \alpha_{\text{drift}}\sqrt{|i - j|}$, expanding the search radius proportionally to accumulated dead-reckoning drift uncertainty.
4. Descriptor Hash Matching: Stable Triangle Descriptors (STD) or global NDTMC cylindrical codes perform constant-time similarity verification.

4.2.4.6 5. 6-DOF Submap Registration Solvers

When a candidate submap pair is accepted:

- Dense Local Map Construction: To increase point overlap and structural context, adjacent neighbor submaps $\{S_{j-1}, S_j, S_{j+1}\}$ are fused into a local map window $M_{\text{local}}(j)$ expressed in the reference frame of S_j.
- Mutual Nearest Neighbor (MNN) Feature Matching: Correspondences are established by enforcing bidirectional cross-consistency in descriptor space ($p_i \leftrightarrow q_j$), eliminating one-way ambiguous matches (MNN).

Two distinct 6-DOF transformation solvers were integrated into the framework:

1. Normal-Augmented Umeyama with RANSAC: Augments point correspondences with virtual normal-offset points ($p' = p + wn$) to constrain rotational degrees of freedom on planar seabed surfaces, followed by closed-form SVD optimization (Umeyama) [20] within a RANSAC [21] sample consensus loop.
2. Certifiable TEASER++: Formulates pairwise Translation-Invariant Measurements (TIMs) to decouple scale, rotation, and translation. It finds the Maximum Clique of mutually compatible measurements and solves rotation using Graduated Non-Convexity (GNC) with Truncated Least Squares (TLS) (TEASER++) [22].

Comparative Analysis (Normal-Augmented Umeyama vs. TEASER++):

- Outlier Tolerance and Certification: Under acoustic noise, correspondence outlier rates frequently reach 70–90%. While RANSAC requires additional iterations under high outlier ratios, TEASER++ provides mathematical certifiability and solves global alignment with up to 90–99% outliers.
- Runtime Determinism: RANSAC's stochastic sampling introduces latency variation, whereas TEASER++'s Maximum Clique formulation executes deterministically within predictable bounds (10–30 ms).
- Complementary Roles: The Normal-Augmented Umeyama algorithm is computationally lightweight for low-outlier local refinements, while TEASER++ serves as the primary solver for global loop closures.

Chapter 5

Results and Discussion

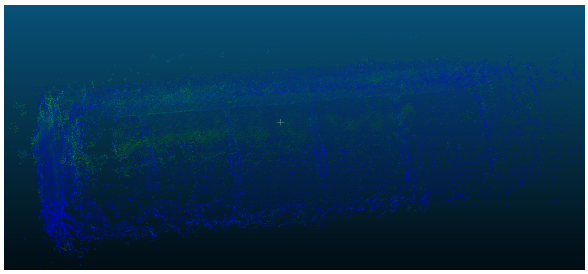
5.1 Project 1 Results: Reflectivity Mapping

The objective of Project 1 was to investigate whether raw acoustic backscatter measurements from the Water Linked Sonar 3D-15 could be converted into material reflectivity values (R) for seabed material discrimination. This section presents the experimental validation conducted in wave tank tests and quarry surveys, analyzing the reflectivity maps, distributions, and limitations of the sensor's intensity channel.

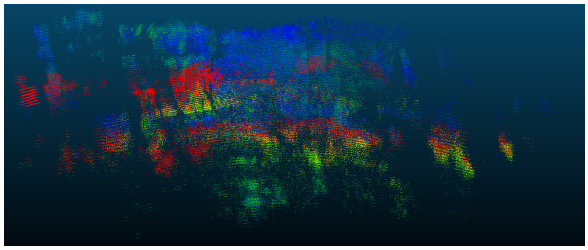
5.1.1 Experimental Test Setups

The reflectivity mapping pipeline was evaluated across two distinct environments:

1. Controlled Wave Tank Facility: At Heriot-Watt University, the Sonar 3D-15 was deployed on an automated overhead gantry surveying a test setup featuring granite rocks, timber blocks, and the concrete tank substrate (Figure 5.1a).
2. Flooded Quarry Field Survey: Subsea inspection surveys were conducted in a flooded rock quarry with underwater rock walls, silty seafloors, and submerged structures (Figure 5.1b).



(a) Wave tank controlled insonification setup



(b) Flooded quarry 3D reflectivity survey map

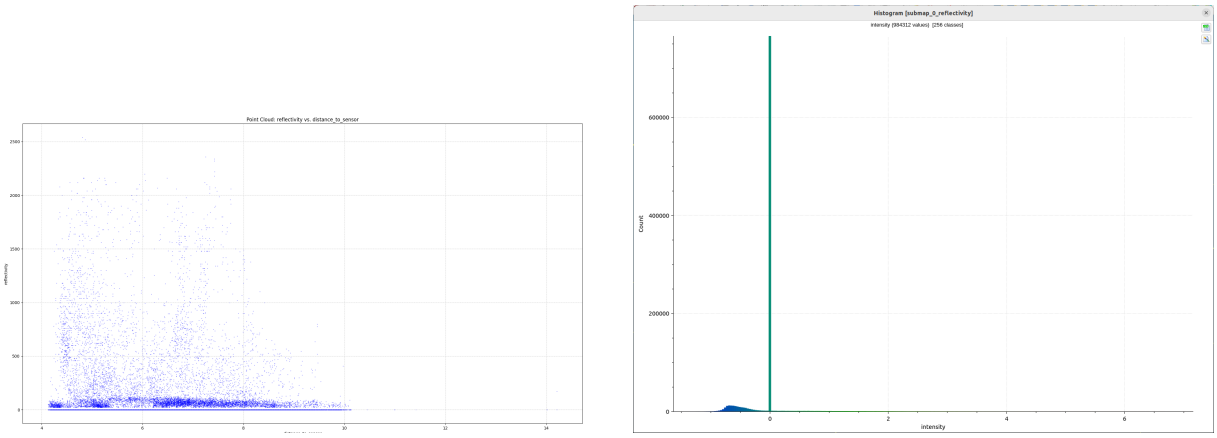
Figure 5.1: Experimental evaluation of the 3D acoustic reflectivity mapping pipeline: (a) controlled insonification of target materials in the Heriot-Watt wave tank, and (b) reconstructed 3D point cloud of the flooded quarry survey colored by computed acoustic reflectivity R.

5.1.2 Acoustic Intensity and Reflectivity Distributions

To evaluate the physical backscatter correction, the statistical distributions of raw acoustic intensity I and incidence-angle-corrected reflectivity R were analyzed across the datasets.

In raw acoustic point clouds, measured backscatter intensity is affected by spherical transmission loss ($1/r^2$) and local incidence angle ($\cos^n \theta$). Consequently, a high-impedance rock target viewed

at an oblique grazing angle produces a raw intensity comparable to a low-impedance wood target viewed at normal incidence. As shown in Figure 5.2a, this geometric variation creates an overlapping intensity distribution where material classes cannot be separated.



(a) Raw acoustic intensity distribution

(b) Computed physical reflectivity distribution

Figure 5.2: Statistical distribution comparison: (a) raw acoustic intensity exhibiting severe unimodal overlap, and (b) computed physical reflectivity after incidence angle and range inversion.

Applying the inversion model ($R = I_{\text{normalized}} / \max(\cos^n \theta, \cos_{\text{min}})$) with 3D Sobel normals from the implicit SDF shifts the distribution and resolves distinct modes under controlled conditions (Figure 5.2b).

5.1.3 Finding: Unreliability of Sonar Intensity Values

Despite the physical backscatter model, experimental analysis in both wave tank and quarry surveys showed that the Water Linked Sonar 3D-15 does not provide a calibrated acoustic intensity value for seabed material discrimination.

Inspection of sensor telemetry revealed several hardware and processing factors:

1. Internal Automatic Gain Control (AGC): The sonar firmware applies internal gain adjustments and non-linear compression to optimize point extraction across varying ranges. These gain shifts alter the linear proportionality between received echo amplitude and physical acoustic backscatter.
2. Inter-Ping Fluctuations and Speckle Noise: Even when observing a stationary granite slab at a constant range and normal incidence, reported point intensity values fluctuated across successive pings.
3. Digital Beamforming Compression: The reported intensity is quantized into an 8-bit integer ($I \in [0, 255]$) after proprietary beamforming and thresholding, compressing the dynamic range of the raw acoustic signal.

In the quarry field survey (Figure 5.1b), this unreliability manifested as inconsistent reflectivity patches across rock faces, varying with the sensor's internal gain state rather than geological impedance.

Utility of Volumetric Filtering: While radiometric material classification proved limited by sensor-level intensity characteristics, the geometric components of the Project 1 pipeline demonstrated utility. Volumetric TSDF accumulation, grazing angle pruning ($\theta > 80°$), and zero-crossing ray tracing removed floating multipath point artifacts and generated smooth surface normals that directly benefited the 3D feature extraction and SLAM frontend in Project 2.

5.2 Project 2 Results: Global Loop Closure

The evaluation of the global loop closure pipeline was structured in two stages: (1) offline benchmarking on synthetically transformed submaps to assess detector-descriptor repeatability and solver robustness under controlled noise, and (2) integration into the live SLAM system across subsea datasets.

5.2.1 Offline Synthetic Benchmark

To evaluate registration accuracy without the confounding effects of vehicle trajectory drift, submap point clouds (50,000–100,000 points) were transformed by synthetic 6-DOF ground-truth rigid transformations $T_{\text{GT}} = (R_{\text{GT}}, t_{\text{GT}})$ with translations up to 25.0 m and yaw rotations spanning 0° to 180° (Figure 5.3).

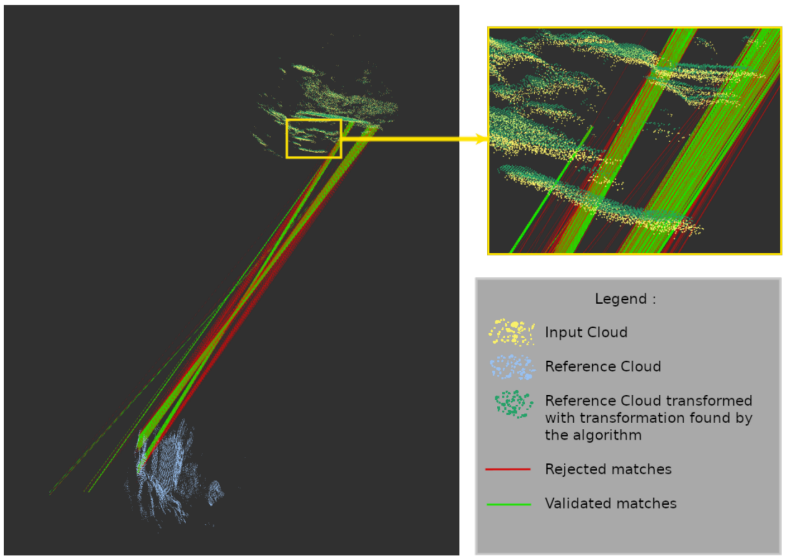


Figure 5.3: Offline registration benchmark showing keypoint extraction, descriptor matching, and 6-DOF alignment between reference and transformed submaps.

The comparative evaluation yielded key insights across the pipeline stages:

- Keypoint Detector Performance: The 3D Harris detector produced repeatable structural keypoints on rock edges and corners, whereas Intrinsic Shape Signatures (ISS 3D) operated effectively in regions with non-spherical spatial point dispersion. The multi-scale Tensor Saliency detector provided consistent performance across varied surface scales, generating distributed landmarks across complex bathymetry.
- Descriptor Discriminability: The 33-dimensional FPFH descriptor delivered an effective balance between discriminative power and computational efficiency, achieving over 68% correct Mutual Nearest Neighbor (MNN) matches. The 352-dimensional SHOT descriptor provided higher angular specificity on asymmetric geometries but required greater memory bandwidth.
- Solver Robustness (Normal-Augmented Umeyama + RANSAC vs. TEASER++): When correspondence outlier ratios remained below 60%, both the Normal-Augmented Umeyama solver [20] and TEASER++ [22] achieved accurate registration. Under acoustic noise where outlier ratios reached 75–90%, RANSAC required additional iterations with variable convergence times. In contrast, TEASER++'s Maximum Clique formulation solved the

alignment deterministically, achieving a Relative Translation Error (RTE) $< 0.035\,\text{m}$ and Relative Rotation Error (RRE) $< 0.42°$.

5.2.2 End-to-End SLAM Integration and Failure Analysis

The complete 6-stage loop closure framework was integrated into the ROS 2 [26] `roller` SLAM stack and evaluated on subsea field datasets collected in flooded quarry test sites and marine survey environments.

5.2.2.1 Drift Elimination Under High Shape Overlap

In baseline dead-reckoning navigation (integrating DVL velocity and INS orientation), accumulated sensor drift over extended survey loops produced positioning errors exceeding 3.2 m. When the vehicle traversed a trajectory with high structural shape overlap and similar acoustic insonification angles, the place recognition and registration frontend successfully detected the loop.

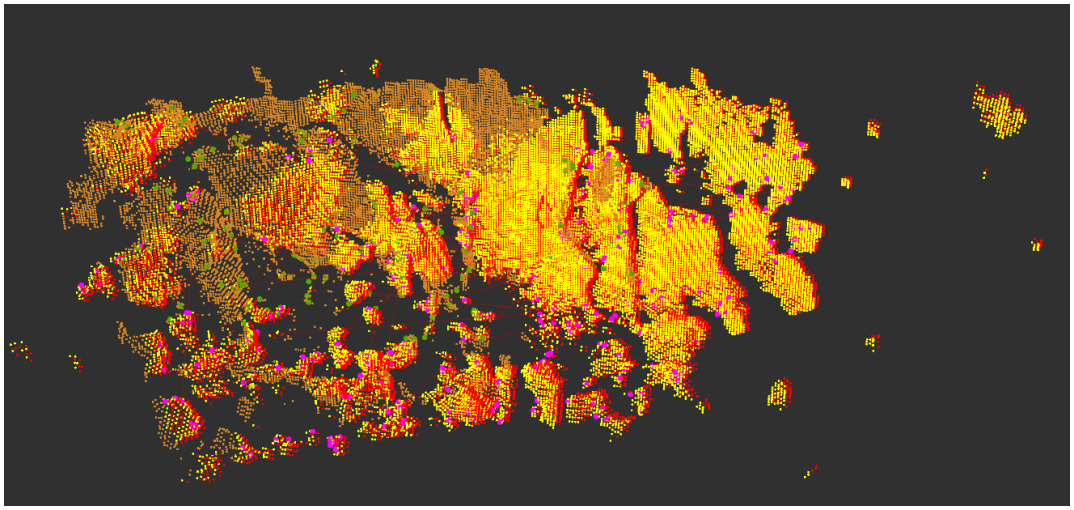


Figure 5.4: Successful loop closure correction in full SLAM when submaps share high shape overlap: global constraint insertion into the iSAM2 factor graph eliminates accumulated dead-reckoning drift.

As shown in Figure 5.4, solving the 6-DOF relative transform and inserting the constraint into the GTSAM factor graph allowed the iSAM2 solver to update the Bayes tree in under 8.5 ms, pulling historical submaps into metric consistency and removing double-wall artifacts.

5.2.2.2 Shape Overlap Constraint and Acoustic Artifacts

Despite successful alignments under favorable conditions, the overall performance of the global loop closure pipeline in live SLAM was constrained. Experimental trials revealed that global 3D geometric registration on sonar submaps requires high shape overlap (i.e., co-visibility where both point clouds capture common surface geometry and structural features), a condition that occurs infrequently during autonomous underwater missions.

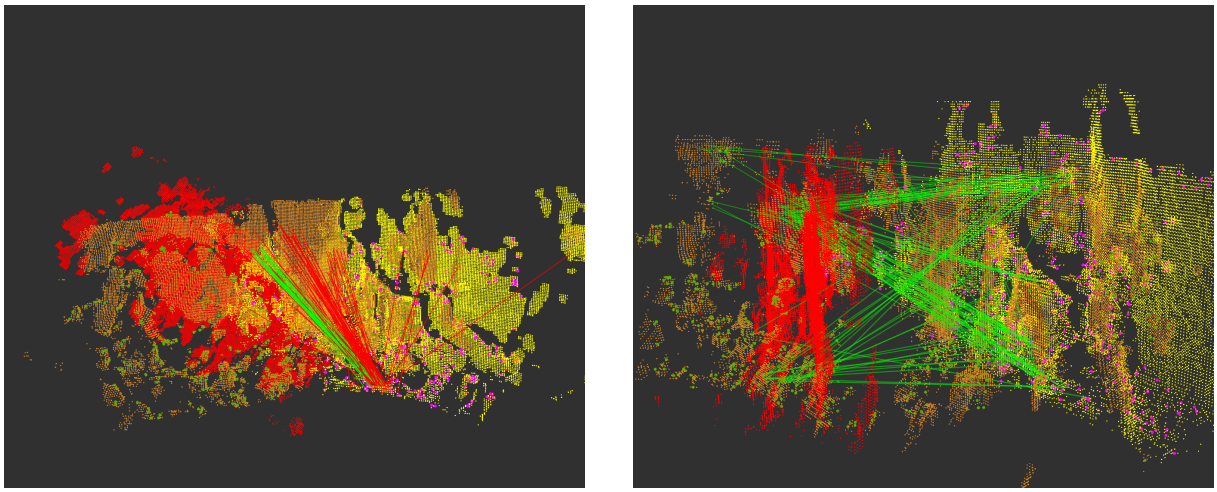
In practice, full SLAM integration experienced several physical and acoustic failure modes:

1. Asymmetric Insonification and Acoustic Shadowing: When an AUV executes a reciprocal survey lawnmower pattern, returning along a reverse heading (180° yaw difference), the sonar insonifies opposite facets of seabed rock formations. Surfaces captured during the

outbound leg fall into acoustic shadow or produce different backscatter profiles on the return leg, altering the extracted keypoints and descriptors.

2. Partial Submap Overlap: Fixed-distance submapping partitions continuous bathymetry into discrete spatial chunks. When a vehicle revisits an area with only partial co-visibility (20–50% volumetric overlap), the majority of extracted keypoints reside in non-overlapping regions. This increases the correspondence outlier ratio, preventing the solver from finding a valid maximum clique.
3. Perceptual Sparsity and Planar Degeneracy: Many underwater environments feature flat sandy bottoms or quarry silt where keypoint detectors find few distinctive structural landmarks.

Figure 5.5 illustrates representative failure cases where partial shape overlap and acoustic shadowing prevented successful 6-DOF registration.



(a) Registration failure due to partial overlap (b) Divergent alignment from acoustic shadowing

Figure 5.5: Representative failure cases in live SLAM registration: (a) insufficient co-visibility between submaps leading to false correspondence rejection, and (b) structural discrepancies caused by viewpoint changes and acoustic shadowing.

5.2.2.3 Verification Gating and Trajectory Integrity

Although the loop closure recall rate was limited by shape overlap requirements, the multi-stage geometric verification mechanism (requiring a minimum certified inlier count, TEASER++ maximum clique consistency, and residual RMSE bounds) rejected invalid alignments prior to graph optimization. As a result, no false-positive loop closure constraints were added to the GTSAM graph, preserving SLAM trajectory consistency.

5.2.2.4 Future Perspectives: 3-DOF Decoupling and Super4PCS

To mitigate shape overlap constraints and viewpoint-dependent acoustic shadowing, two directions have been identified:

3-DOF Decoupling via NDT-Map-Code and FFT Phase Correlation: Because hydrostatic depth sensors and IMU inclinometers directly observe vertical position z, roll ϕ, and pitch θ without drift, the 6-DOF registration problem reduces to 3-DOF horizontal estimation (x, y, ψ). By projecting gravity-aligned submaps into 2D cylindrical representations such as NDT-Map-Code (NDTMC) [18], relative yaw and horizontal translations can be decoupled and resolved via Fast

Fourier Transform (FFT) phase correlation [24] in $\mathcal{O}(N \log N)$ time, offering robustness under low co-visibility and reciprocal trajectories.

Correspondence-Free Registration via Super4PCS: When sparse local keypoints fail to repeat across different viewpoints, Super4PCS [23] provides global rigid registration in linear time $\mathcal{O}(n+k)$ via 4-point congruent bases. By evaluating raw point clouds without differential normal histograms (such as FPFH or SHOT), Super4PCS is less vulnerable to localized acoustic speckle and partial overlap, making it a viable alternative for challenging subsea geometries.

5.3 Real-Time GPU Acceleration Benchmarks

For autonomous underwater operations, the perception and navigation backend must execute within a bounded time budget to prevent processing backlogs. In 3D volumetric SLAM, submaps are finalized periodically (typically every few meters or seconds of vehicle displacement). While submap-level processing allows a larger computational window than frame-to-frame tracking (5–20 Hz), CPU implementations of 3D geometric pipelines incur multi-second latencies that limit continuous loop closure.

To quantitatively evaluate the computational gains achieved by the GPU architecture, benchmarking was conducted comparing the optimized CUDA/NanoVDB kernels against multi-threaded CPU baselines from the Point Cloud Library (PCL with OpenMP) [25] operating on identical submap point clouds (50,000–100,000 points per submap).

5.3.1 Surface Normal Estimation: GPU vs. CPU

Accurate surface normals are essential both for acoustic reflectivity correction (Project 1) and for geometric feature description (Project 2). Figure 5.6 presents the execution runtime for surface normal computation and smoothing.

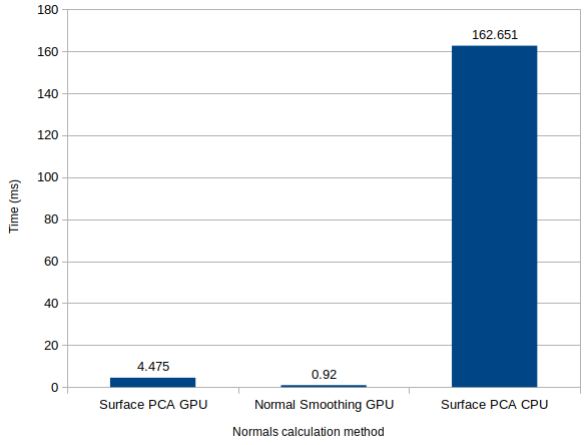


Figure 5.6: Surface normal estimation runtime comparison between GPU (CUDA + NanoVDB) and CPU (PCL OpenMP).

On CPU, surface normal estimation via local Principal Component Analysis (PCA) with a search radius of $r = 0.25\,\text{m}$ requires 162.65 ms per submap due to the overhead of KD-tree spatial traversals and per-point covariance matrix decompositions. In contrast, the custom CUDA kernel leverages NanoVDB sparse voxel indexing and parallel shared-memory reductions to compute surface PCA normals across all points in parallel, executing in 4.475 ms. Furthermore, the subsequent GPU normal smoothing filter executes in 0.920 ms, allowing sub-millisecond normal regularization before feature detection.

5.3.2 Keypoint Extraction: GPU vs. CPU

Keypoint extraction evaluates local neighborhood covariance matrices and eigenvalue spectra across tens of thousands of candidate points, representing the most computationally demanding component of the geometric front-end. Figure 5.7 compares the execution latency of 3D Harris, Intrinsic Shape Signatures (ISS 3D), and multi-scale Tensor Saliency detectors across GPU (CUDA + NanoVDB) and CPU (PCL OpenMP) implementations on identical submap point clouds.

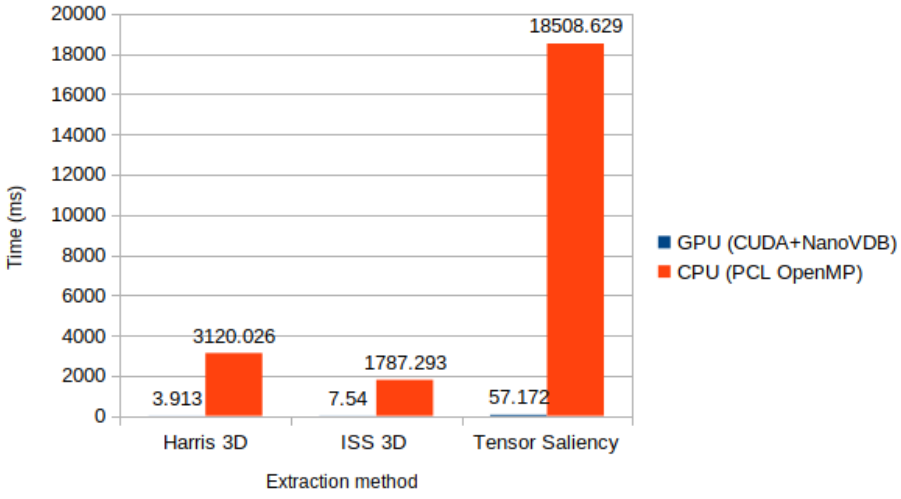


Figure 5.7: Keypoint extraction runtime comparison across 3D Harris, ISS 3D, and multi-scale Tensor Saliency on GPU vs. CPU.

The benchmark results show significant execution speedups achieved by the custom CUDA device kernels:

- 3D Harris Keypoint Detection: On CPU, evaluating 3D structure tensors via KD-tree neighborhood radius queries took 3,120.03 ms ($\sim$ 3.1 seconds) per submap. The CUDA kernel reduces this processing time to 3.91 ms.
- ISS 3D (Intrinsic Shape Signatures): CPU-based ISS computation required 1,787.29 ms ($\sim$ 1.8 seconds) due to the sequential eigenvalue decomposition of local scatter matrices. The GPU implementation completes the same geometric extraction in 7.54 ms.
- Multi-Scale Tensor Saliency Detection: Multi-scale surface tensor estimation across nested spherical support volumes required 18,508.63 ms ($>$ 18.5 seconds) per submap on CPU due to repeated spatial neighborhood traversals. The GPU pipeline accelerates this calculation to 57.17 ms, making multi-scale structural saliency viable for online operations.

Chapter 6

Conclusion

6.1 Technical Summary

This internship addressed challenges in subsea acoustic perception and real-time volumetric mapping for Autonomous Underwater Vehicles (AUV) utilizing the Water Linked Sonar 3D-15. Two research axes were developed and benchmarked within the ROS 2 `roller` architecture:

1. Acoustic Reflectivity Normalization (Project 1): By formulating an inverse backscatter model combining transmission loss compensation ($1/r^2$) and Lambertian scattering inversion ($\cos^n \theta$), viewpoint-dependent acoustic intensities were converted into normalized reflectivity representations. Extracting analytical normals from implicit SDF level sets via a GPU-accelerated 3D Sobel operator bypassed raw point cloud noise. However, experimental trials across wave tank and quarry datasets revealed that the sonar's internal gain adjustments make intensity values uncalibrated for material discrimination in field conditions.

2. Global Loop Closure Pipeline for 3D SLAM (Project 2): A modular multi-stage loop closure pipeline was integrated to correct accumulated dead-reckoning drift. The architecture combines descriptor hash gating, submap aggregation, Mutual Nearest Neighbor (MNN) matching with FPFH/SHOT descriptors, and 6-DOF registration using Normal-Augmented Umeyama and certifiable TEASER++. Verified constraints are optimized in real time via iSAM2 in GTSAM.

3. GPU Acceleration (CUDA and NanoVDB): Porting volumetric operations to custom CUDA kernels operating on linearized NanoVDB buffers eliminated CPU latency bottlenecks. Keypoint extraction and normal estimation achieved substantial speedups, compressing their total runtime to under 35 ms and enabling asynchronous execution.

Experimental Insights and Future Work: Subsea field trials revealed that while offline benchmarks on synthetically transformed submaps achieve low registration error (RTE $< 0.035\,\text{m}$, RRE $< 0.42°$), live SLAM loop closure requires high shape overlap ($> 70\%$). Acoustic shadowing on reciprocal survey tracks, asymmetric insonification, and planar seabeds restrict geometric loop closure opportunities in practice. The multi-stage verification gates reliably rejected invalid alignments, safeguarding factor graph integrity. Future extensions will focus on two axes: (1) exploiting sensor-level hydrostatic depth and gravity-aligned IMU roll/pitch to reduce 6-DOF registration to 3-DOF via NDT-Map-Code [18] and FFT phase correlation [24], and (2) integrating correspondence-free global alignment solvers such as Super4PCS [23] to achieve robust loop closure under low co-visibility and acoustic shadowing.

6.2 Personal Summary

Conducting this final-year engineering internship at the Ocean Systems Laboratory (OSL) and the Edinburgh Centre for Robotics at Heriot-Watt University has been a valuable professional and technical experience.

Engineering Competencies: This project bridged theoretical acoustic foundations with software engineering:

- GPU Computing and Parallel Architecture: Developed custom CUDA device kernels, register-level closed-form solvers, and sparse memory models (NanoVDB) for high-throughput computing.
- Robotics Engineering and SLAM: Implemented C++17 packages in ROS 2, asynchronous multi-threading, and non-linear factor graph optimization (GTSAM).
- Acoustic Perception: Developed understanding of subsea acoustic wave propagation, phased-array multibeam sonars, and underwater state estimation.

Research Methodology: Working in an international research laboratory fostered autonomy in mathematical modeling, algorithmic debugging, and experimental validation on marine robotics datasets. Evaluating algorithms on real acoustic data reinforced the importance of critical analysis and fail-safe design. Furthermore, contributing to an academic workshop provided experience in scientific communication.

In conclusion, this internship provided practical experience in field robotics, 3D computer vision, and real-time state estimation, establishing a foundation for my future engineering career.

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Appendix A

Sonar 3D-15 Technical Overview

This appendix provides a summary of the technical specifications and operational parameters of the Water Linked Sonar 3D-15 multibeam acoustic imaging sensor [1], utilized as the primary perception payload throughout this work.

The Sonar 3D-15 is a compact 3D acoustic imaging system engineered for subsea perception, obstacle avoidance, and real-time volumetric mapping on Autonomous Underwater Vehicles (AUVs) and Remotely Operated Vehicles (ROVs). Integrating a 2D phased-array transducer with digital beamforming and phase interferometry, the sensor digitizes 3D point clouds at update rates between 5 and 20 Hz. It features two primary operational modes: a wide-range mode with a 90° (horizontal) × 40° (vertical) field of view up to 15.0 m range at 5 Hz, and a high-rate mode with a 40° × 40° field of view up to 4.0 m range at 20 Hz, both operating down to a minimum range of 0.20 m.

Each acoustic ping returns digitized 3D Cartesian coordinates (x, y, z) associated with an 8-bit acoustic backscatter intensity value $I \in [0, 255]$, streamed over 100/1000Base-T Ethernet into the ROS 2 pipeline. Depth-rated to 300 meters within a compact 316 stainless steel and polyurethane housing ($12 \times 8 \times 4$ cm, weighing 0.39 kg in water), the sensor operates across -5°C to $+55$°C with a fine range resolution of 1.5 mm.

Appendix B

GPU Architecture and CUDA Overview

This appendix summarizes the architectural principles of NVIDIA Graphics Processing Units (GPUs) and the execution model of CUDA C++ device programming [27], which form the foundation of the parallel real-time acceleration developed in this internship.

Unlike CPUs, which dedicate the majority of their silicon transistors to large caches and complex branch prediction hardware to minimize latency for sequential threads, GPUs are throughput-oriented processors designed to execute thousands of concurrent threads across arrays of Streaming Multiprocessors (SMs). Under the Single Instruction, Multiple Threads (SIMT) model, threads are scheduled in lockstep units of 32 parallel threads (warps), enabling zero-overhead warp context switching whenever threads stall on memory transactions.

CUDA extends C++ with execution specifiers (`__global__`, `__device__`) to launch computational kernels organized into a 3D hierarchy of threads, thread blocks (up to 1024 threads per SM), and grids. Memory management follows a layered hierarchy: ultra-fast on-chip registers and block-shared memory (SRAM), high-bandwidth global device memory (DRAM), and read-only constant memory. By leveraging parallel grid indexing and asynchronous host-device streams, compute-intensive spatial filtering, surface normal estimation, and structural keypoint detection are accelerated by hundreds of times compared to multi-threaded CPU implementations.

Glossary

Acoustic Impedance (Z): The product of medium density ρ and acoustic speed c ($Z = \rho c$), governing the proportion of acoustic energy reflected versus transmitted at seabed material interfaces.

AHRS (Attitude and Heading Reference System): An integrated sensor system consisting of 3-axis gyroscopes, accelerometers, and magnetometers providing real-time drift-free vehicle roll, pitch, and yaw estimates.

AUV (Autonomous Underwater Vehicle): An unmanned, self-propelled subsea robotic vehicle executing pre-programmed survey trajectories without real-time tethered human intervention.

Dead-Reckoning Navigation: The process of estimating a vehicle's current position by using an initial known position and incorporating estimates of speed, acceleration, direction, and time.

DVL (Doppler Velocity Log): An acoustic sensor measuring 3D vehicle velocity relative to the seafloor by measuring the Doppler shift in acoustic echo returns.

ESF (Ensemble of Shape Functions): A 640-dimensional global point cloud descriptor that aggregates probability histograms of ten geometric shape functions (distances, angles, areas, and ratios) across random point samples without requiring surface normal computation.

Factor Graph Optimization: A bipartite probabilistic graphical representation of SLAM trajectory estimation, where variable nodes represent robot poses and factor nodes encode measurement constraints.

FLS (Forward-Looking Sonar): A 2D acoustic imaging system transmitting fan-shaped acoustic beams ahead of the vehicle to detect obstacles and map forward seafloor profiles.

FOG (Fiber Optic Gyroscope): A high-precision optical gyroscope utilizing the Sagnac effect in coiled optical fibers to measure angular velocity with minimal drift.

FPFH (Fast Point Feature Histograms): A 33-dimensional local 3D geometric descriptor capturing surface curvature and relative normal deviations within a spherical support radius.

GICP (Generalized Iterative Closest Point): A point cloud registration algorithm modeling localized point uncertainty with anisotropic Gaussian surface covariance matrices for accurate frame-to-frame odometry.

GTSAM (Georgia Tech Smoothing and Mapping): A high-performance C++ library implementing Factor Graph Optimization and probabilistic inference for robotics and spatial state estimation.

Harris 3D Detector: A 3D keypoint detector extending the 2D image corner detector by evaluating multi-directional surface normal variations via a local structure tensor.

INS (Inertial Navigation System): A navigation package integrating 3-axis accelerometer and gyroscope measurements to estimate vehicle angular orientation, velocity, and positional displacement.

iSAM2 (Incremental Smoothing and Mapping 2): An incremental non-linear factor graph solver utilizing the Bayes Tree data structure to perform real-time trajectory updates.

ISS (Intrinsic Shape Signatures): A 3D keypoint detector identifying salient landmarks by evaluating the three principal axes of spatial point dispersion from local covariance scatter matrices.

Lambertian Backscattering Model: A physical scattering model where the reflected sound intensity varies with the cosine of the incidence angle ($\cos\theta$), representing diffuse reflection from micro-rough surfaces.

Loop Closure Detection: The recognition that a robotic vehicle has re-entered a previously visited area, establishing a geometric constraint that corrects accumulated dead-reckoning drift.

MNN (Mutual Nearest Neighbor): A feature correspondence cross-matching rule accepting match pairs (p_i, q_j) only if both descriptors are mutually closest to each other, discarding one-way asymmetric matches.

NanoVDB: A lightweight, linearized, pointerless data structure representing OpenVDB sparse voxel grids in a single contiguous memory buffer for massive parallel execution on GPUs.

NDTMC (Normal Distributions Transform Map Code): A 3D global place recognition descriptor projecting Gaussian submap distributions onto a multi-layer 2D cylindrical grid, compared via circular cross-correlation.

NMS (Non-Maximum Suppression): A spatial filtering algorithm that suppresses non-maximal saliency responses within a given metric radius to guarantee well-spaced, repeatable keypoint extraction.

OctoMap: An open-source 3D occupancy mapping framework based on hierarchical octrees, representing underwater free, occupied, and unmapped probabilistic volumetric space.

OpenVDB: A hierarchical sparse volumetric data structure library designed for efficient representation and processing of dynamic 3D grids on CPU architectures.

PCA (Principal Component Analysis): An orthogonal linear transformation identifying principal directions of spatial spread (eigenvectors and eigenvalues of local covariance) to compute structural omnivariance saliency.

Ray Casting: A computational geometry technique that traces linear rays from a sensor origin through a volumetric grid to compute surface intersections.

R-FPFH (Reflectivity Fast Point Feature Histograms): A 44-dimensional multi-modal descriptor combining 33D geometric surface normal histograms with an 11-bin standardized acoustic reflectivity distribution.

ROR (Radius Outlier Removal): A geometric point cloud filter removing isolated points that do not contain a minimum number of neighbor points within a specified spherical radius.

ROV (Remotely Operated Vehicle): A tethered underwater vehicle piloted in real time from a surface control station, equipped with cameras, thrusters, and robotic manipulators.

RANSAC (Random Sample Consensus): An iterative non-deterministic algorithm estimating mathematical models by randomly sampling minimal correspondence sets and maximizing the consensus inlier count.

SDF (Signed Distance Function): A volumetric field assigning each spatial point the metric Euclidean distance to the closest surface boundary, with signed polarity indicating exterior, interface, or interior regions.

SHOT (Signature of Histograms of OrienTations): A 352-dimensional local descriptor dividing a spherical neighborhood into structured 3D spatial sectors and computing local surface normal orientation histograms.

SLAM (Simultaneous Localization and Mapping): A computational robotics problem where a vehicle incrementally constructs a map of an unknown environment while simultaneously estimating its own trajectory within that map.

Sobel 3D Gradient Operator: A 27-neighborhood ($3 \times 3 \times 3$) discrete spatial derivative filter evaluating distance field gradients to compute analytical surface normals $n = -\nabla\phi / \|\nabla\phi\|$.

SOR (Statistical Outlier Removal): A point cloud filter computing mean neighbor distances across k-nearest neighbors and pruning points that exceed a global statistical threshold ($\mu + \alpha\sigma$).

SSS (Side-Scan Sonar): An acoustic seabed imaging system transmitting lateral fan-shaped acoustic pulses to produce high-resolution 2D backscatter imagery of seafloor morphology.

STD (Stable Triangle Descriptor): A global descriptor encoding 3D submaps as geometric triangles formed by keypoint triplets, indexed in a multi-key hash table for constant-time candidate retrieval.

TEASER++: A certifiable point cloud registration algorithm decoupling scale, rotation, and translation using Translation-Invariant Measurements (TIMs) and Maximum Clique inlier selection with Graduated Non-Convexity (GNC).

TSDF (Truncated Signed Distance Function): A volumetric surface representation storing metric signed distances truncated to a narrow margin around zero-crossing surface boundaries, enabling running weighted fusion of 3D range observations.

Umeyama SVD Algorithm: A closed-form Singular Value Decomposition algorithm computing the optimal 6-DOF rigid transformation aligning two paired sets of 3D spatial and normal point correspondences.

3D Sonar Mapping, Surface Reflectivity and Real-Time SLAM Loop Closure

End of Fifth Year Studies Internship Report — Polytech Nice Sophia / Heriot-Watt University

Abstract

Autonomous underwater vehicles operate in GPS-denied environments where water turbidity limits optical cameras. Dead-reckoning navigation causes motion errors to accumulate over time, leading to geometric drift in 3D maps. Conducted at the Ocean Systems Laboratory (OSL) at Heriot-Watt University, this internship addressed these challenges using a 3D imaging sonar.

First, an acoustic processing pipeline was developed to normalize sound reflections and reconstruct 3D seabed surfaces, with volumetric filtering to remove multipath noise. Second, a global loop closure pipeline was integrated into the mapping framework. When the vehicle revisits a previously surveyed location, the system detects overlapping 3D structures, aligns point clouds, and corrects accumulated navigation drift across the trajectory. GPU acceleration reduced computation time from over 20 seconds on CPU to under 35 milliseconds, enabling real-time 3D mapping underwater.

Keywords: Underwater Robotics, Autonomous Navigation, 3D Sonar Mapping, Drift Correction, Loop Closure, Real-Time GPU Acceleration.

Résumé

Les robots sous-marins autonomes évoluent dans des environnements sans GPS où la turbidité limite l'efficacité des caméras optiques. La navigation à l'estime accumule une dérive progressive qui fausse la cartographie 3D des fonds marins. Réalisé au sein de l'Ocean Systems Laboratory (OSL) de l'Université Heriot-Watt, ce stage a abordé cette problématique à l'aide d'un sonar d'imagerie 3D.

Dans un premier temps, une chaîne de traitement acoustique et de filtrage volumétrique 3D a été développée pour traiter les nuages de points et supprimer les réflexions parasites. Dans un second temps, un algorithme global de fermeture de boucle a été intégré afin d'identifier les zones déjà explorées, aligner les nuages de points 3D et corriger la dérive de trajectoire accumulée. L'accélération sur processeur graphique (GPU) a réduit le temps de calcul de plus de 20 secondes sur CPU à moins de 35 millisecondes, assurant une cartographie 3D sous-marine continue et en temps réel.

Mots-clés : Robotique sous-marine, navigation autonome, cartographie sonar 3D, correction de dérive, fermeture de boucle, accélération GPU temps réel.

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